

SAMREEN KAUR

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Summary

Fourth-year Science student at the University of British Columbia with hands-on experience in software development, application testing, API validation, database analysis, and data analytics through professional and academic projects. Proficient in Python, Java, SQL, C++, and R, with experience using Oracle SQL, Azure, OpenShift and GitHub Actions in technical environments. Demonstrated ability to analyze complex systems, validate data accuracy, debug automated tests, and build data-driven solutions. Strong problem-solving, communication, and teamwork skills with experience working in collaborative technical environments. Passionate about applying technical and analytical skills to solve complex problems and develop effective solutions. Seeking software engineering, QA automation, or data-focused opportunities where strong analytical and technical skills can drive measurable impact.

Skills

- **Programming:** Python, Java, C++, C, SQL, R, JavaScript, HTML, CSS
- **Software & Development Tools:** Git, Jira, Confluence, IntelliJ IDEA, Jupyter, RStudio, Linux, Windows, macOS
- **Cloud & Development Platforms:** Azure, AWS, IBM Cloud, Google Cloud, GitHub Actions, OpenShift (OCP)
- **Data & Analytics:** Oracle SQL, Apache Spark, TensorFlow, Pandas, NumPy, MATLAB, Data Analysis, Statistical Analysis, Machine Learning, Deep Learning, Data Visualization
- **Technical Skills:** Data Structures & Algorithms, Software Testing, API Testing, Regression Testing, Database Validation, Batch Processing, Debugging
- **Interpersonal:** Problem Solving, Critical Thinking, Teamwork, Time Management, Written & Verbal Communication

Education

- University of British Columbia, Vancouver, BC
Bachelor of Science
Expected graduation: December 2026

Certification

- Professional Certificate
IBM Data Science - Advanced Python
- Specialization
HTML, CSS, and JavaScript for Web Developers - John Hopkins

Professional Experience

Software/Application Developer Co-op - CIBC
Toronto, ON

May 2026 – August 2026

- Performed production and system validation for Private Wealth applications by reconciling API/XML responses and account-level totals against expected results, identifying discrepancies and data anomalies across DIT and SIT testing cycles.
- Executed 20+ batch testing scenarios across multiple environments using Git, Azure, OpenShift, and GitHub Actions, generating input files and validating outputs to ensure accuracy, consistency, and system stability.
- Debugged Java-based automated test scripts and executed API regression test cases during an ESB modernization SIT certification cycle, validating services supporting account information, trade confirmations, tax receipts, and report retrieval.
- Wrote and executed Oracle SQL queries to verify backend database updates, confirm successful record creation, and reconcile application results during application and batch testing.
- Created and tracked Jira tickets for defects, test scenarios, and validation activities, while documenting technical workflows and batch processes in Confluence to maintain testing traceability and support knowledge sharing.

**Data Analyst – Data Science UBC
Vancouver, BC**

May 2023 – Aug 2023

- Collected and analyzed data from SQL databases to support development of a sustainable campus project.
- Used Python and R for data processing, analysis, and development of a working analytical design.
- Collaborated with a data science research team to investigate and communicate data-driven findings.

Technical Projects

Exoplanet Transit Signal Classification

Jan 2026 – Apr 2026

- Trained and compared five machine learning models (Logistic Regression, kNN, Decision Trees, Random Forest, SVM) to classify exoplanet transit signals from NASA TESS data.
- Addressed severe class imbalance using precision, recall, and F1-score, and optimized a Random Forest via grid search with cross-validation.
- Result: achieved 93.9% accuracy and 0.95 AUC, with transit depth, orbital period, and duration identified as the most predictive features.

Interactive Data Visualization Website

Sept 2025 – Dec 2025

- Designed and developed an interactive data visualization website to analyze demographic patterns in billionaire wealth accumulation using a large, multi-source dataset.
- Developed linked and parameterized visualizations (scatterplots, heatmaps, density and violin plots, etc) using Altair (python) to support exploratory analysis.

Traffic Accident Project

May 2025 – Aug 2025

- Analyzed 10 years of open-sourced traffic accident data using Apache Spark.
- Trained machine learning models to classify accident severity; achieved 22% higher F1-score than baseline.
- Deployed model with Google cloud + dashboard for real-time scenario testing

Used car Selling Price Analysis

Sept 2024 – Jan 2025

- Computed covariates having significant effect on the selling price of a used car using R
- Fitted a regression model, used exhaustive selection model to select reduced models of various sizes, visualized the result
- Result: The variables associated with selling price of a used car are the present price, the year, the number of previous owners, the seller type and the mileage of the vehicle.

Predictive model - International student Proportion

Jan 2024 – May 2024

- Verified whether the variables chosen to rank the universities to create an accurate model be used to predict the proportion of international students using R and SQL
- Used forward selection algorithm for predictive analysis, made a RMSE reduced model for accurate results
- Results: forward selection ad correlation coefficient analysis conducted on the training set suggested that the predictive model works best with four input variables.

Additional Experience

Tutor high school students – Coding and Mathematics
Orientation leader – Science UBC
Customer Service Associate – Save on Foods
Science Undergraduate Society UBC – Associate leader

Jan 2023 – Aug 2023
Aug 2023 – Oct2023
July 2023 – Dec 2024
Oct 2022